



Workshop Deep**BLUES**

Deep Learning for Balancing, Linkage and Effects of Selection

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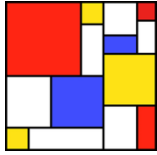
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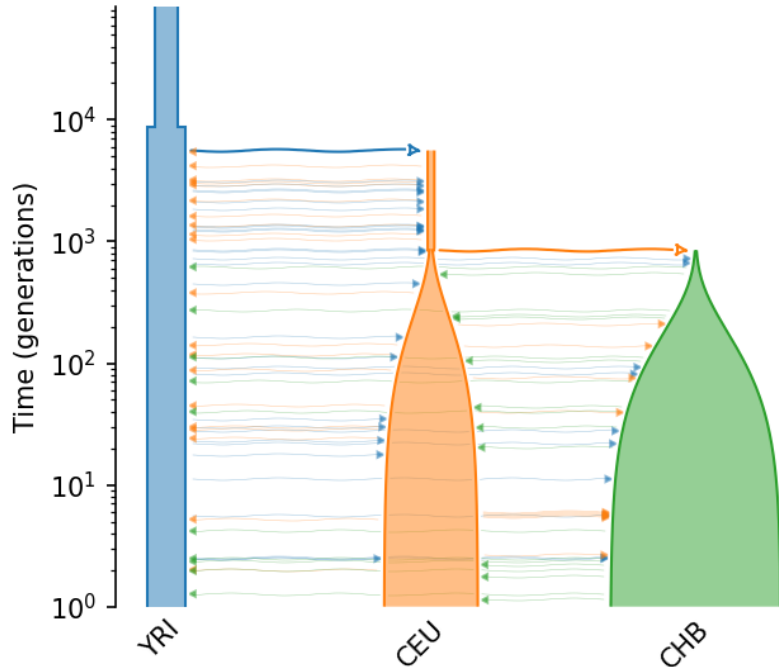
Background

- Balancing selection (BS): long-standing evolutionary puzzle.
 - Drives genetic variation across extended periods of time.
- Detection of BS challenging.
 - Often entangled with Linkage Disequilibrium (LD) effects.
 - Impact of LD on BS detection still inadequately explored.
- Goal is to simulate genomic segments that combine both effects and, through a ML approach, classify the presence of BS.

Simulations



OutOfAfrica_3G09



- Demographic model adapted to our scenarios
 - Genetic map based on GRCh38 genome assembly
- Model parameters were edited in order to introduce variability to the simulations.
 - Selection coefficients
 - Epoch lengths
 - Population numbers at the start of each epoch
 - Migration rates
 - Population growth rates
- 5000 simulations per scenario per gene
 - 500 individuals sampled

SLIM: Messer and Haller (2023) American Naturalist; PopSim: Adrion et al., (2020) and Lauterbur et al., (2023), eLife; Out of Africa model: Gutenkunst et al., (2009) PLoS Genetics

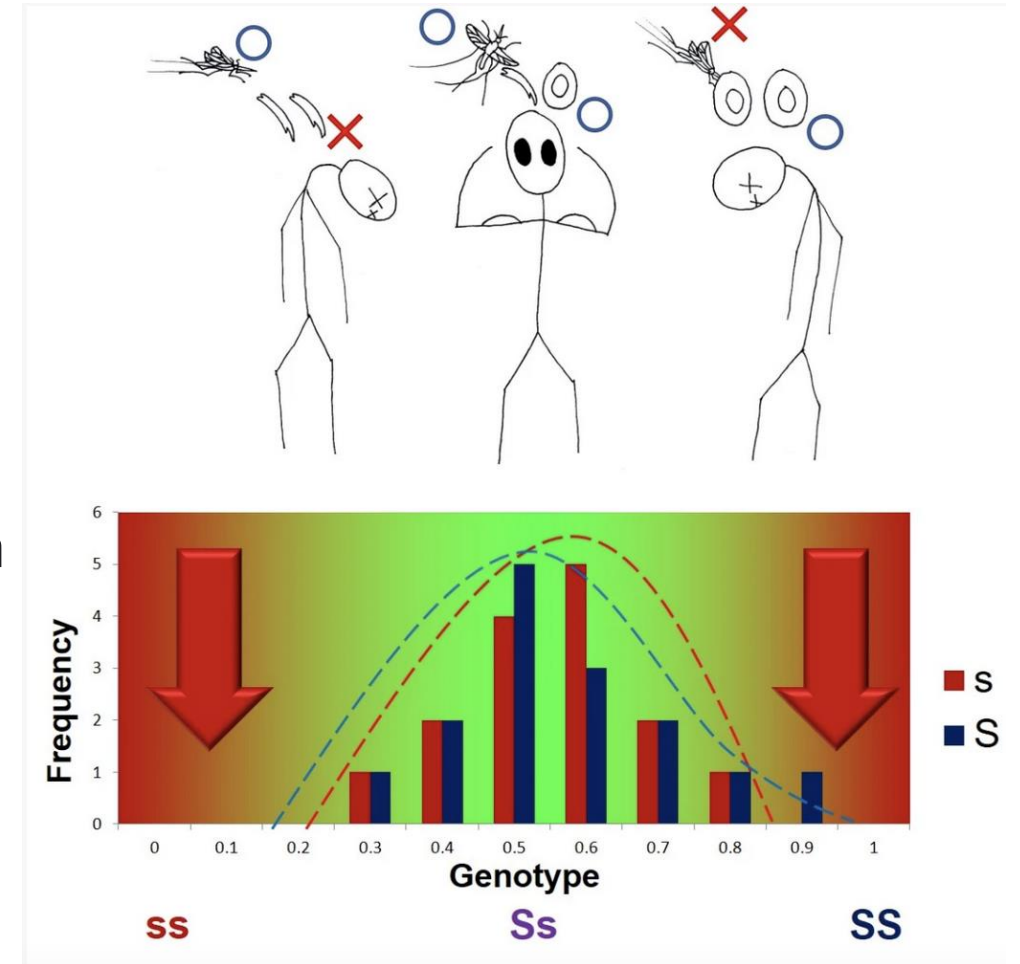
Simulations

- Two distinct scenarios simulated:
 - Neutral scenario: genomic region is allowed to evolve neutrally, with no selective pressure acting upon it.
 - BS scenario: genomic region simulated under BS (heterozygote advantage).
- Genomic regions simulated - HLA-A and HLA-DOA genes:
 - Regions differ in their recombination rates which allows to test how differences in LD could affect detection of BS.

Balancing selection

- Rare alleles are maintained in the population
 - sex-related genes (Connallon and Clark, 2014; Mank, 2017)
 - Sickle-cell disease and immune response in humans (Kelley et al., 2005)
 - pathogen resistance (Bakker et al., 2006)
 - plant and fungi self-incompatibility (Castric and Vekemans, 2004)

heterozygote advantage



Implementing ML framework

- Using population genetics summary statistics extracted from VCF files, and tree sequence data summary statistics from Ancestral Recombination Graphs (ARGs), we are going to use ML to classify different scenarios.
 - Aim is to detect instances of BS.
- Feel free to explore the Jupyter Notebook
 - PopGen_SumStats_markdown_2025Workshop.ipynb, but...

Before we start:

- Create and activate a virtual environment:

```
conda create -n myenv python=3.12.11  
conda activate myenv
```

- Install Jupyter and ipykernel on your newly created environment and register:

```
pip install jupyter ipykernel  
python -m ipykernel install --user --name=myenv --display-name "Python (myenv)"
```

- Install the packages needed for this workshop:

```
pip install ete3==3.1.3 pandas==2.3.0 numpy==2.1.3 scipy==1.15.3  
scikit-learn==1.7.0 tensorflow==2.19.0 joblib==1.5.1 h5py==3.14.0  
Keras==3.10.0 matplotlib==3.10.3 seaborn==0.13.2
```

- Launch Jupyter Notebook, ensuring you select your new kernel.

Population Genetics summary statistics

- Python package scikit-allel used to extract several population genetics summary statistics computed from VCF files.
 - Some statistics were computed directly from code created by Isildak et al., (2021)
- Goal of this first part will be to use a deep learning framework to detect signals of BS using these statistics.

Haplotype diversity statistics

- H1/H12/H123
- H2/H1
- Mean iHS
- Mean nS_L
- Mean EHH decay
- Haplotype diversity

Allele frequency statistics

- Tajima's D
- Fu and Li's D^* and F^*
- Fay and Wu's H
- Zeng's E
- Watterson's θ
- NCD1

Genetic diversity and heterozygosity

- Nucleotide diversity
- Mean pairwise difference
- Mean observed and expected heterozygosity

Linkage disequilibrium and recombination

- Kelly's Z_{ns}
- Mean LD

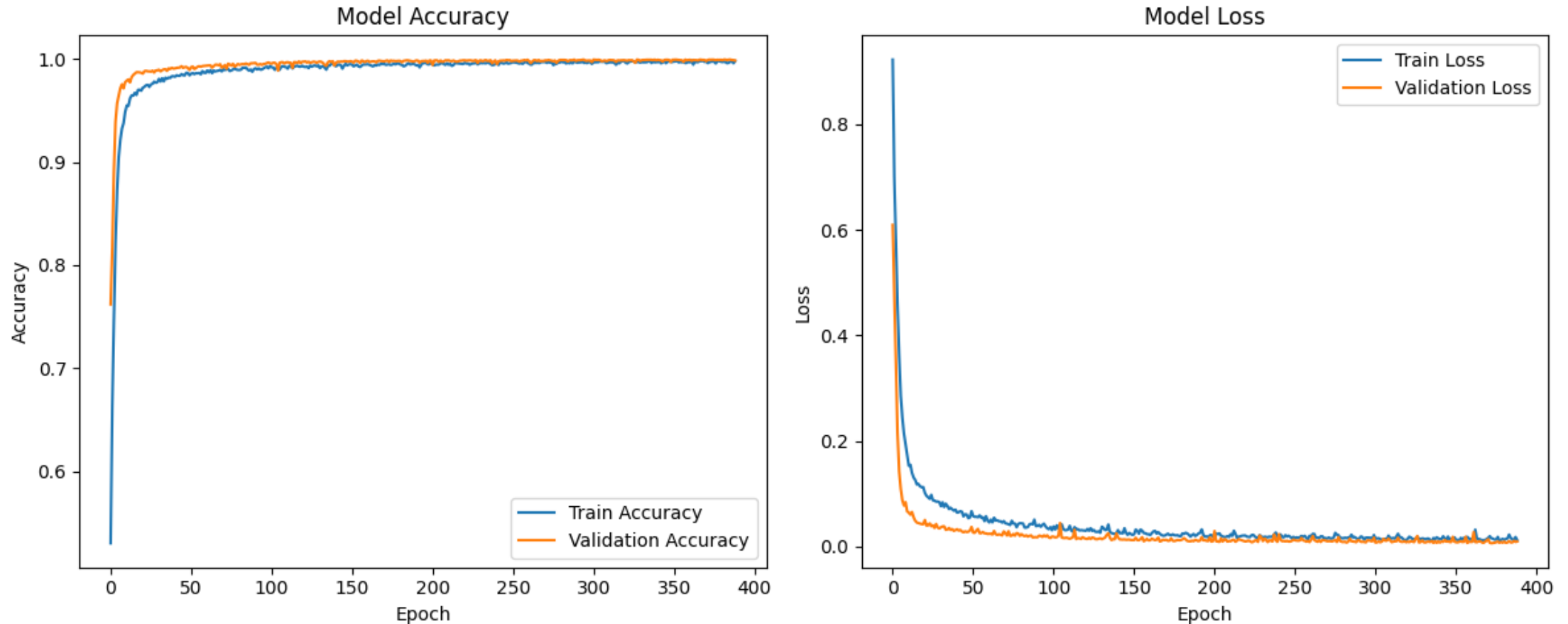
Population structure

- Mean F_{st}
- Raggedness

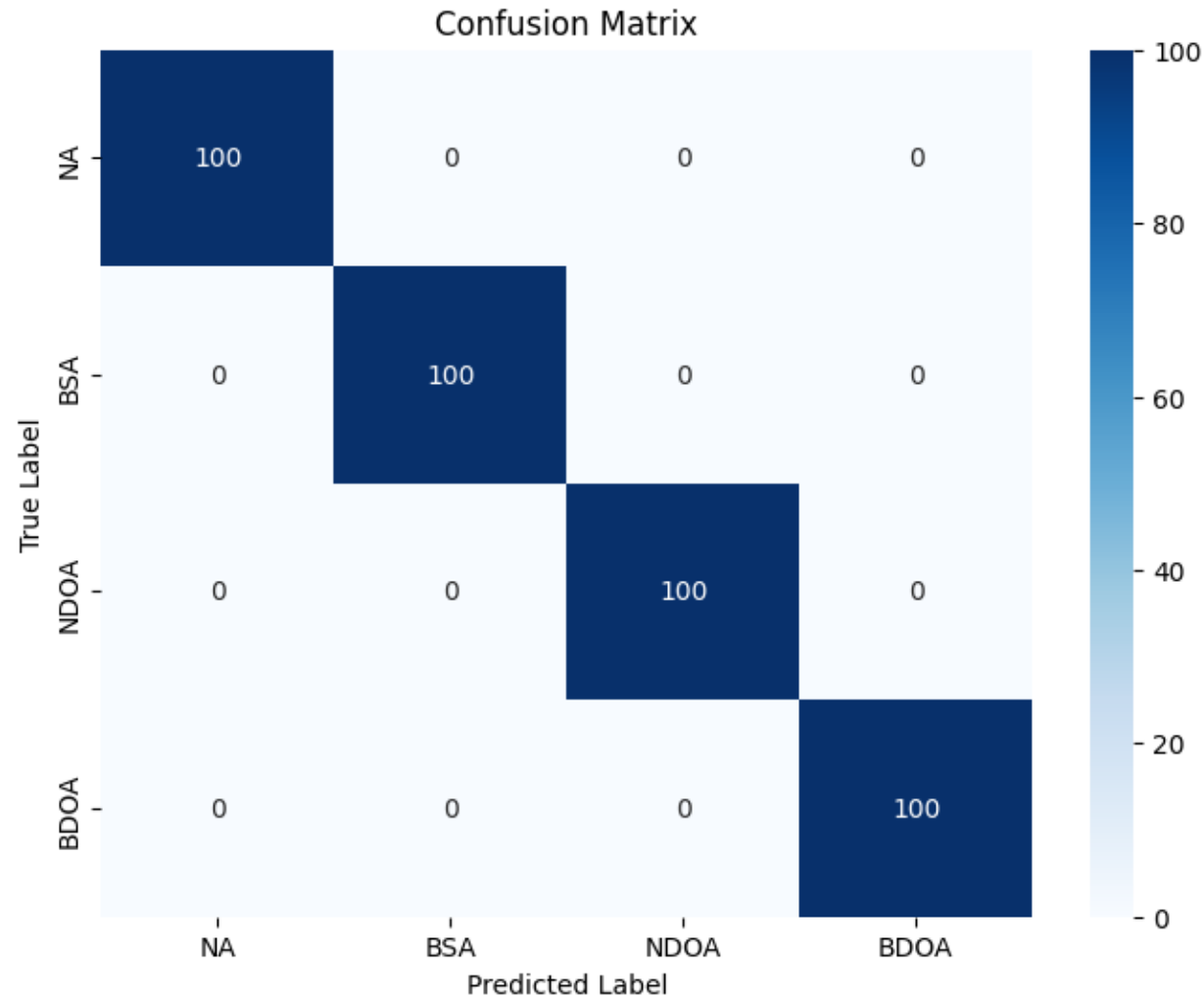
Workflow

- Import required packages
- Read in data
 - Pre-processing steps will transform data into something that can be used as input in this machine learning framework.
- Build model
- Initialize and train the model.

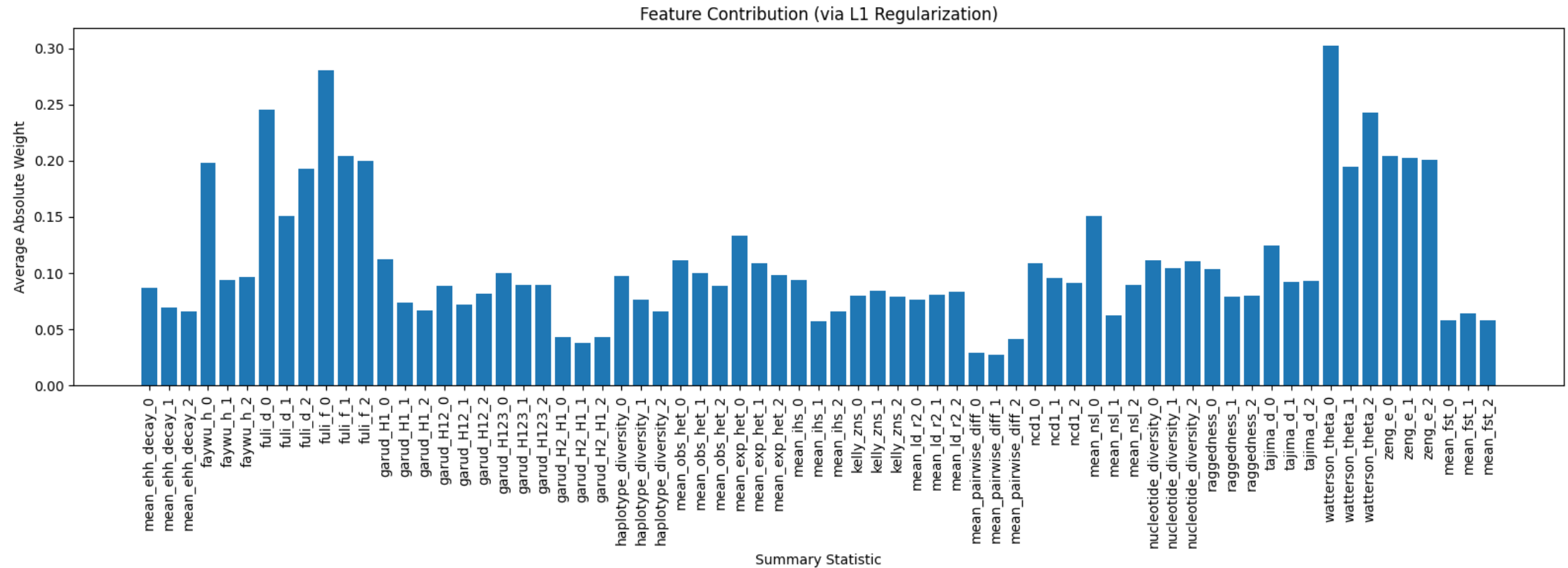
Visualizing the training of the NN



Loading model and assessing performance



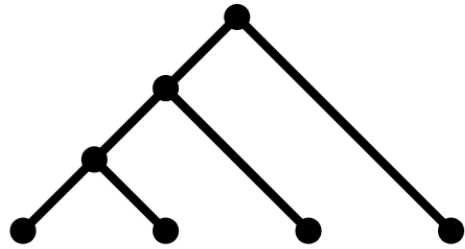
Feature contribution



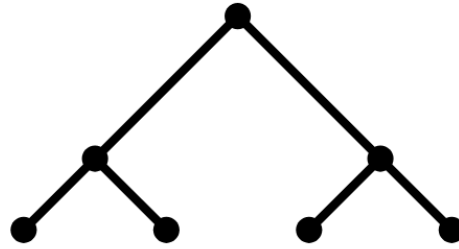
Tree summary statistics

- Tree Shape (esp. Tree Balance Indices)
- Branch Lengths
- Lineages-through-time (LTT)
- All computed using code from Lambert et al. 2023 and the ete3 python toolkit

Tree Balance Indices



Imbalanced



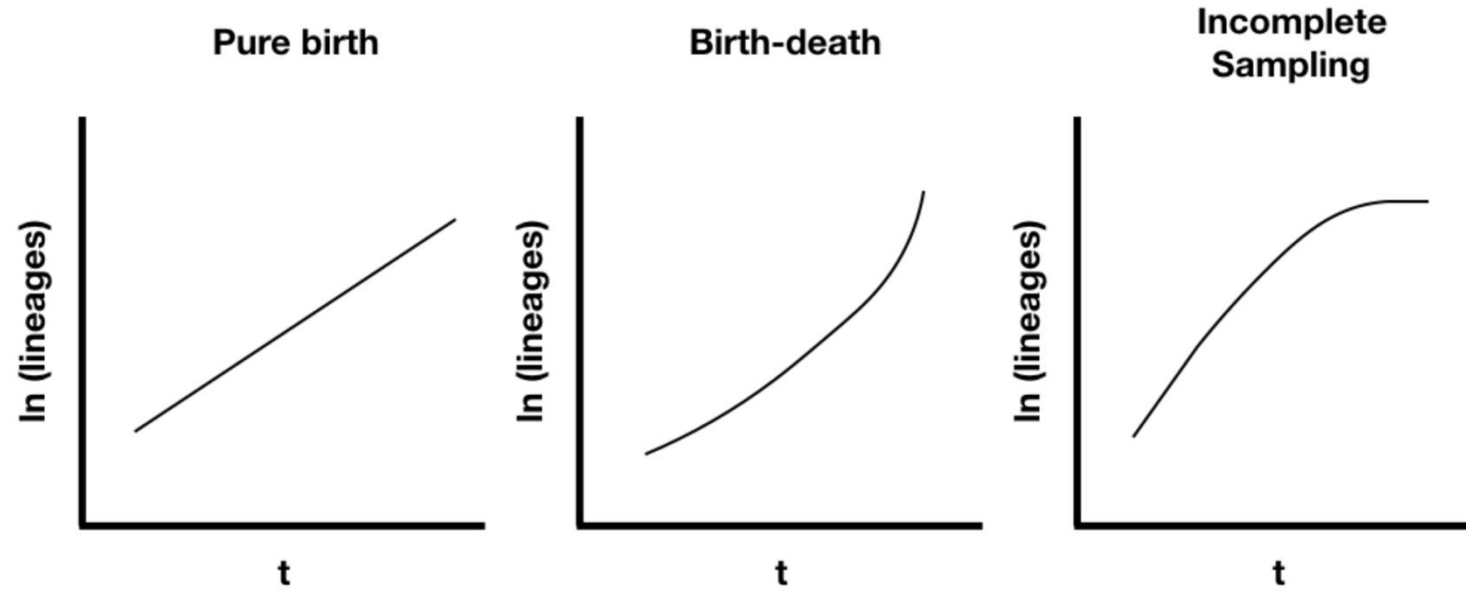
Balanced

- Balance of a tree is a measurable quantity
- Multiple statistics have been invented to do this
- Most famous may be the Sackin and Colless Indices
- Multiple such statistics are computed by our scripts

Branch Length statistics

- A tree is composed of multiple branches with distinct lengths
- Take median, mean, and variance of branch lengths
- Internal, external, altogether

Lineage-through-time analysis

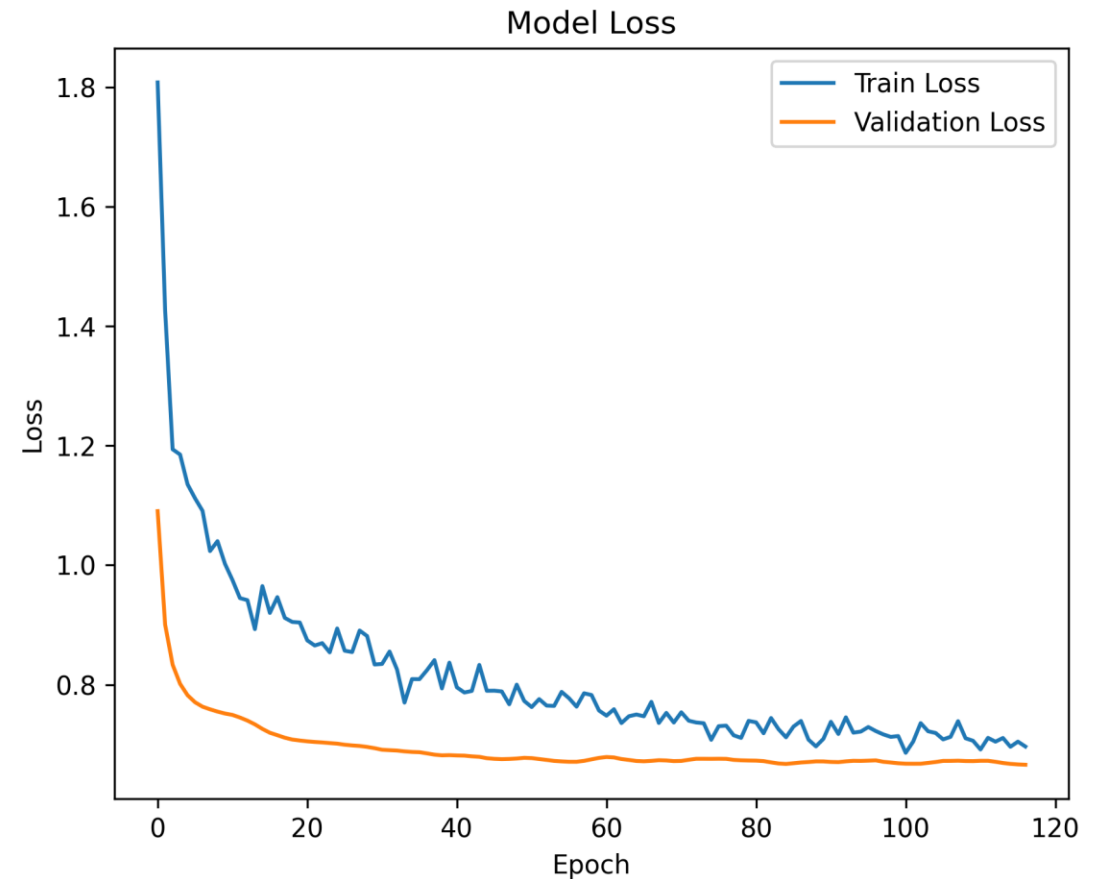
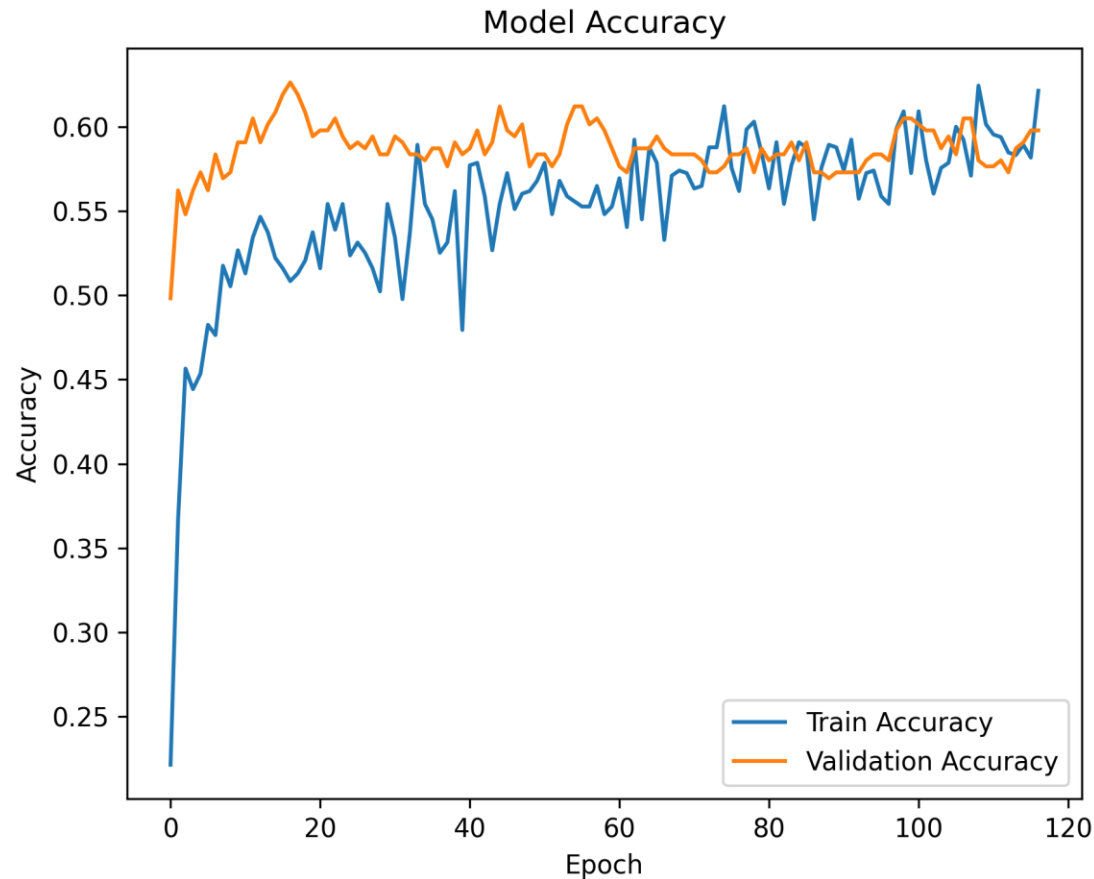


- Lineages generally increase over time due to speciation
- Rate of lineage increase changes due to birth-death processes
- Number of lineages through time is informative of birth-death processes

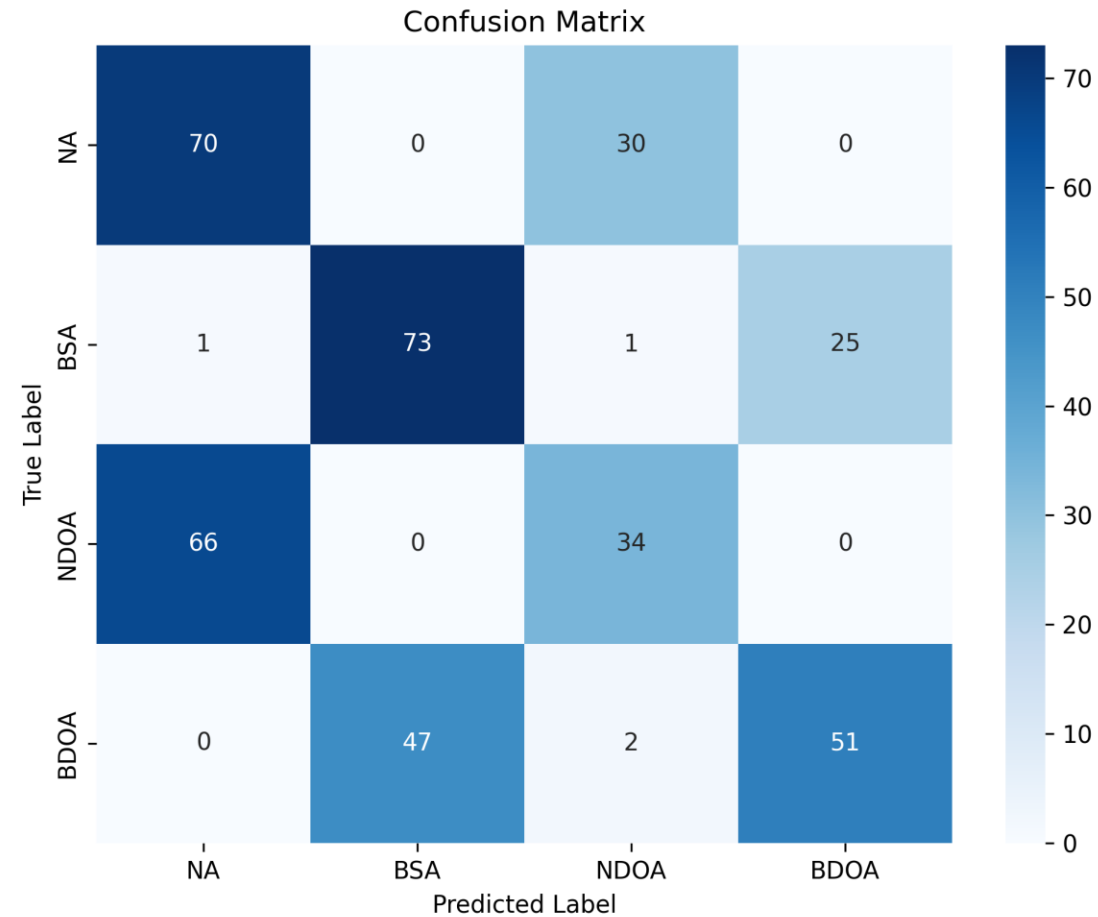
Running Phylogenetic summary statistics

- We have extracted trees in newick format from ARGs
- Each newick file corresponds to a gene and a scenario
 - Balancing HLA-A
 - Neutral HLA-A
 - Balancing HLA-DOA
 - Neutral HLA-DOA
- Begin running the jupyter notebook first
 - SumStats_Phylo_Calculation_Workshop.ipynb for summary stats
 - **IMPORTANT: Statistics take about 30 minutes to compute in total so run all parts of the script and read over while running**
 - SuSt_Phylo_DeepBLUES_Workshop for ML training on summary stats

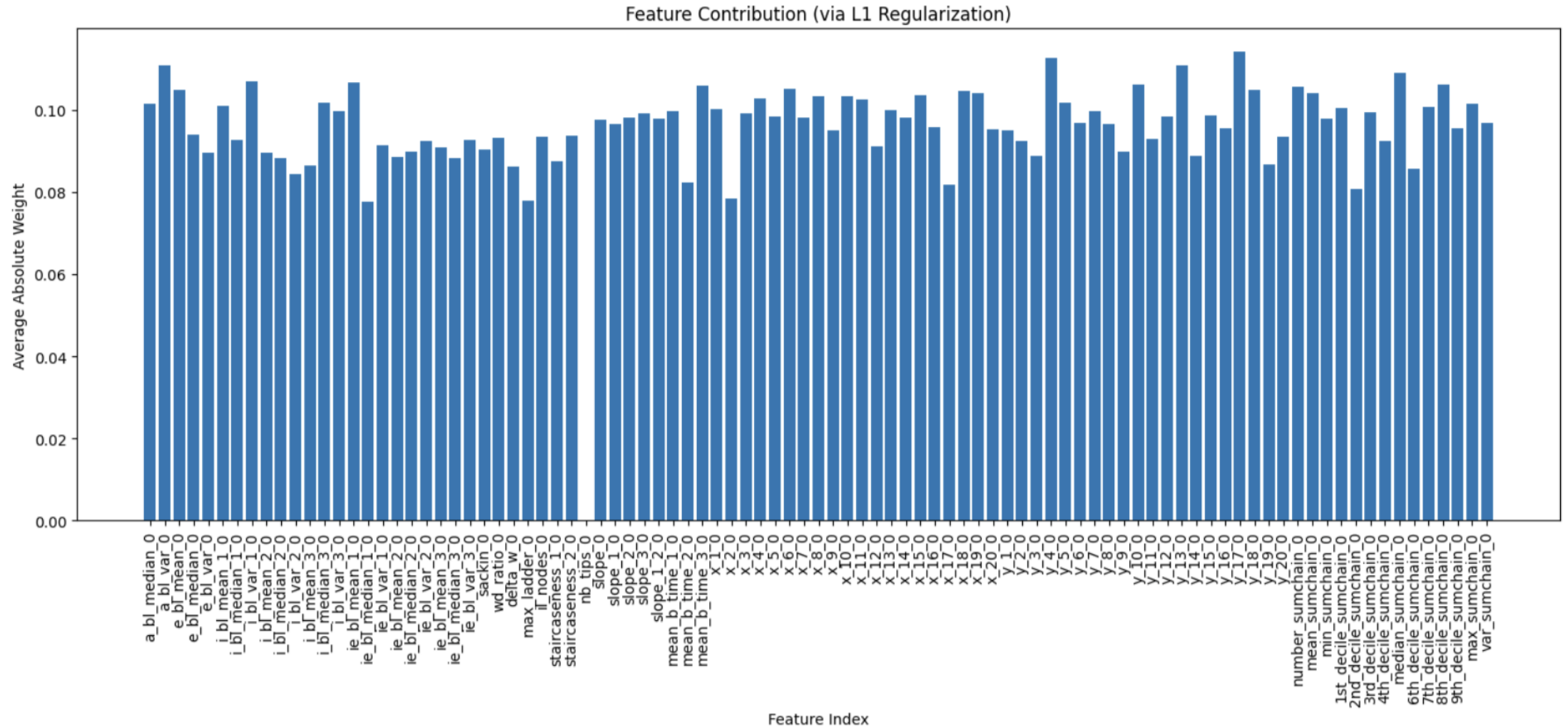
Visualizing the training of the NN



Loading model and assessing performance



Feature contribution



Summary

- Population genetics summary statistics have demonstrated extraordinary accuracy classifying different scenarios.
 - Though not much different, feature contribution suggests frequency-based statistics are slightly more informative.
- Summary statistics from ARGs appear to perform reasonably well.
 - Feature contribution is relatively uniform for ARG summary stats.
- We are working towards developing more complex, trans-species scenarios.
- Ultimate goal is software development, shifting focus from classification to parameter estimation.

DeepBLUEs Project Partners:



- Carolin Kosiol
- Nick Bailey
- Antonio Pacheco
 - University of St Andrews
- Svitlana Braichenko
 - University of Edinburgh
- Adam Siepel
 - Cold Spring Harbor Laboratories, US
- Olivier Gascuel
 - CNRS, France
- Manolo Perez
 - Real Jardín Botánico, CSIC, Madrid, Spain